GitHub: https://github.com/Manichandan12/Machine-Learning-and-Neural-Networks

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An Analysis of Credit Card Customer Segmentation Using Unsupervised Clustering Methods

1.Introduction:

Understanding customer behavior is essential in the credit card industry, where large volumes of data are generated daily. Analyzing this data helps identify patterns that support better decision-making and improved services.

This study focuses on customer segmentation using unsupervised machine learning techniques to group customers based on their financial behavior. The dataset is preprocessed through cleaning, scaling, and dimensionality reduction using PCA.

Clustering algorithms such as KMeans, GMM, and DBSCAN are applied to identify distinct customer segments. The results provide valuable insights for applications like targeted marketing, customer retention, and risk assessment.

2.Why K-means?

K-Means clustering is a widely used unsupervised machine learning algorithm designed to partition a dataset into a predefined number of clusters based on similarity among data points. The objective of the algorithm is to group observations in such a way that data points within the same cluster are more similar to each other than to those in other clusters.

The algorithm operates by minimizing the within-cluster sum of squares (WCSS), which measures the variance within each cluster. This is achieved through an iterative optimization process that adjusts cluster assignments and centroid positions until convergence is reached.

3.Focus of This Study

This study aims to explore and apply unsupervised machine learning techniques to perform customer segmentation on credit card data. The primary focus is on identifying underlying patterns within the dataset and grouping customers based on similarities in their financial behaviour.

The analysis emphasizes the use of multiple clustering algorithms, including K-Means, Gaussian Mixture Models (GMM), and DBSCAN, in order to compare their effectiveness in capturing the structure of the data. Each method offers a different perspective on clustering, allowing for a more comprehensive understanding of how customers can be segmented.

In addition to model implementation, this study focuses on essential preprocessing steps such as data cleaning, feature scaling, and dimensionality reduction using Principal Component Analysis (PCA). These steps ensure that the dataset is well-prepared for clustering and that the results are both meaningful and reliable.

Another key aspect of this work is the evaluation and interpretation of clustering results. Various performance metrics are used to assess the quality of the clusters, while visualizations and statistical summaries are employed to better understand the characteristics of each segment.

Overall, the focus of this study is not only to apply clustering techniques, but also to interpret the results in a way that provides practical insights. The findings aim to demonstrate how unsupervised learning can be used to transform raw financial data into valuable information for decision-making and customer analysis.

4.Dataset Description

The dataset used in this study contains information related to credit card customers and their financial behaviour. It includes a variety of features that capture customer activity, spending patterns, credit usage, and payment behaviour over a given period.

Each record in the dataset represents an individual customer, while the columns correspond to different attributes that describe their interaction with credit card services. These features provide valuable insights into how customers utilize their credit cards and manage their finances.

Key Features

The dataset typically includes variables such as:

- Balance: The remaining balance on the customer's credit account
- Purchases: Total amount spent using the credit card
- Credit Limit: The maximum credit available to the customer
- Payments: Total payments made by the customer
- Cash Advances: Amount withdrawn as cash using the credit card
- Transaction Frequency: Frequency of purchases or transactions
- Instalment Purchases: Payments made in instalments

These features collectively help in understanding different aspects of customer behaviour, including spending habits, repayment patterns, and credit utilization.

Data Characteristics

- The dataset consists of numerical variables suitable for clustering analysis
- Some features may have different scales, requiring normalization or standardization
- Missing values may be present and are handled during preprocessing
- The dataset does not contain predefined labels, making it suitable for unsupervised learning.

5.0 Data Preprocessing

Data preprocessing is a crucial step in the analytical workflow, as it ensures that the dataset is clean, consistent, and suitable for machine learning algorithms. In this study, several preprocessing techniques are applied to prepare the credit card dataset for clustering analysis.

5.1 Handling Missing Values

The dataset is first examined for missing values across all features. Any incomplete records are removed to maintain data integrity and avoid potential bias in the clustering process. This step ensures that the algorithms operate on a complete and reliable dataset.

5.2 Feature Scaling

Since the dataset contains variables measured on different scales (e.g., transaction amounts, frequencies, and credit limits), feature scaling is performed to standardize the data. The StandardScaler technique is used to transform the features so that they have a mean of zero and a standard deviation of one. This is particularly important for distance-based algorithms such as K-Means and DBSCAN, as it prevents features with larger magnitudes from dominating the clustering process.

5.3 Dimensionality Reduction

To simplify the dataset and improve computational efficiency, Principal Component Analysis (PCA) is applied. PCA reduces the number of features by transforming the original variables into a smaller set of uncorrelated components that retain most of the variance in the data. In this study, the dataset is reduced to two principal components, which also facilitates visualization of clusters in a two-dimensional space.

5.4 Data Transformation

The pre-processed and scaled data is converted into a structured format suitable for modelling. The transformed dataset is then used as input for the clustering algorithms, ensuring consistency across all methods applied in the study.

5.5 Importance of Preprocessing

These preprocessing steps play a vital role in improving the performance and accuracy of clustering algorithms. By ensuring that the data is clean, normalized, and well-structured, the analysis can produce more meaningful and interpretable results.

6.0 Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) is an essential step in understanding the structure, patterns, and relationships within a dataset before applying machine learning techniques. In this study, EDA is performed to gain insights into the distribution of features, detect anomalies, and identify trends in customer behaviour.

6.1 Overview of the Data

The dataset consists of numerical variables representing various aspects of credit card usage, including spending behaviour, transaction frequency, and credit utilization. Initial inspection of the dataset helps in understanding the range, scale, and variability of each feature.

6.2 Distribution Analysis

To examine how individual features are distributed, statistical summaries and visualizations such as histograms are used. This helps identify skewness, outliers, and variations in customer behaviour. For instance, certain features like purchases or cash advances may show a right-skewed distribution, indicating that a small number of customers account for higher values.

6.3 Correlation Analysis

A correlation matrix is used to analyze relationships between features. This helps identify strongly related variables, which may influence clustering results. Visual tools such as heatmaps provide a clear representation of these relationships, highlighting both positive and negative correlations among variables.

6.4 Detection of Outliers

Outliers are identified through visualizations such as box plots. These extreme values can significantly affect clustering algorithms, especially those based on distance measures. Understanding the presence of outliers helps in making informed decisions during preprocessing and model selection.

6.5 Feature Relationships

Scatter plots and pairwise comparisons are used to explore relationships between variables. These visualizations provide insight into how different features interact and whether natural groupings exist within the data.

6.6 Key Observations

EDA reveals important characteristics of the dataset, such as variability in customer spending patterns, differences in credit usage, and potential correlations between financial attributes. These observations guide the selection of appropriate preprocessing techniques and clustering algorithms.

7.0 Model Building

In this study, multiple unsupervised machine learning algorithms are applied to segment credit card customers based on their behavioural and financial attributes. The use of different clustering techniques allows for a comprehensive analysis of the dataset, as each method captures distinct structural characteristics of the data.

7.1 KMeans Clustering

KMeans is implemented as a baseline clustering method due to its simplicity and efficiency. The algorithm partitions the dataset into a predefined number of clusters by minimizing the within-cluster variance.

To determine the optimal number of clusters, the Elbow Method is employed by analysing the within-cluster sum of squares (WCSS) across different values of K. Additionally, Silhouette Analysis is used to evaluate the quality of clustering and validate the chosen number of clusters.

Further refinement is achieved through hyperparameter tuning using GridSearchCV, where parameters such as initialization method, number of initializations, and maximum iterations are

optimized. The final model is trained on the scaled dataset, and cluster labels are assigned to each data point for further analysis and interpretation.

8.0 Gaussian Mixture Model (GMM)

The Gaussian Mixture Model is applied to capture more flexible and probabilistic cluster structures. Unlike KMeans, which assigns each data point to a single cluster, GMM estimates the probability that a data point belongs to each cluster based on underlying Gaussian distributions.

Model selection is performed using the Bayesian Information Criterion (BIC), which balances model fit with complexity. Different combinations of the number of components and covariance types (full, tied, diagonal, and spherical) are evaluated to identify the optimal configuration.

Once the best model is selected, it is fitted to the data, and cluster labels are predicted. This approach allows for better handling of overlapping clusters and provides a more nuanced understanding of customer segmentation.

9.0 DBSCAN

DBSCAN (Density-Based Spatial Clustering of Applications with Noise) is used as a density-based clustering technique that identifies clusters based on the density of data points. Unlike KMeans and GMM, DBSCAN does not require the number of clusters to be specified in advance.

The algorithm relies on two key parameters:

- Epsilon (eps): Defines the radius within which points are considered neighbors
- Minimum samples (min_samples): Specifies the minimum number of points required to form a dense region

Hyperparameter tuning is conducted using GridSearchCV to identify the optimal values for these parameters. DBSCAN is particularly effective in detecting clusters with irregular shapes and identifying noise or outliers, which are labeled separately.

10.0 Summary

By applying KMeans, GMM, and DBSCAN, this study leverages multiple clustering approaches to gain a well-rounded understanding of the dataset. Each algorithm contributes unique insights, enabling a more robust and comprehensive customer segmentation analysis.

11.Results and Discussion

This section presents the outcomes of the clustering models applied to the dataset, along with visualizations to support interpretation and comparison.

11.1 K-Means Clustering Results

The KMeans algorithm was applied after determining the optimal number of clusters using the Elbow Method. The results indicate that the dataset can be effectively segmented into distinct groups based on customer behaviour.

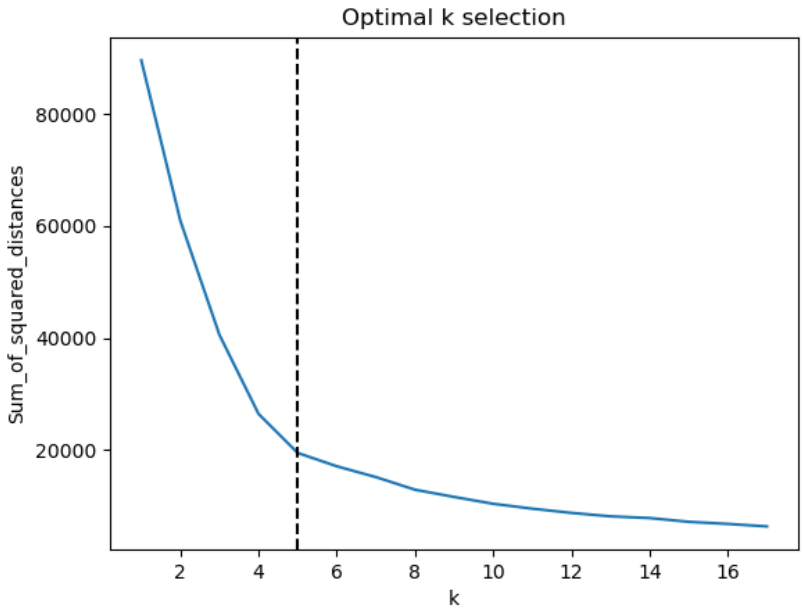

Figure: Elbow plot showing the optimal number of clusters based on inertia.

The elbow point suggests the optimal number of clusters, beyond which the improvement in clustering performance becomes marginal.

11.2 KMeans Cluster Visualization :

The plot shows clear separation between clusters, indicating that KMeans effectively captures the underling structure of the data.

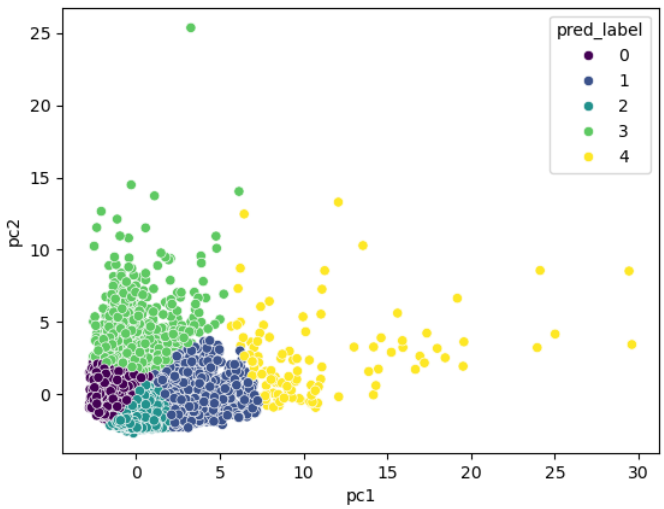

Figure: Visualization of KMeans clusters in two-dimensional PCA space.

12.0 Silhouette Analysis

K=3:

The silhouette plot for k = 3 shows an average score of 0.48, indicating moderate clustering quality. Most data points have positive silhouette values, suggesting they are well assigned to their clusters.

Cluster 1 appears well-defined, while Clusters 0 and 2 show some variability. Overall, the clusters are reasonably separated, making k = 3 a suitable choice for segmentation.

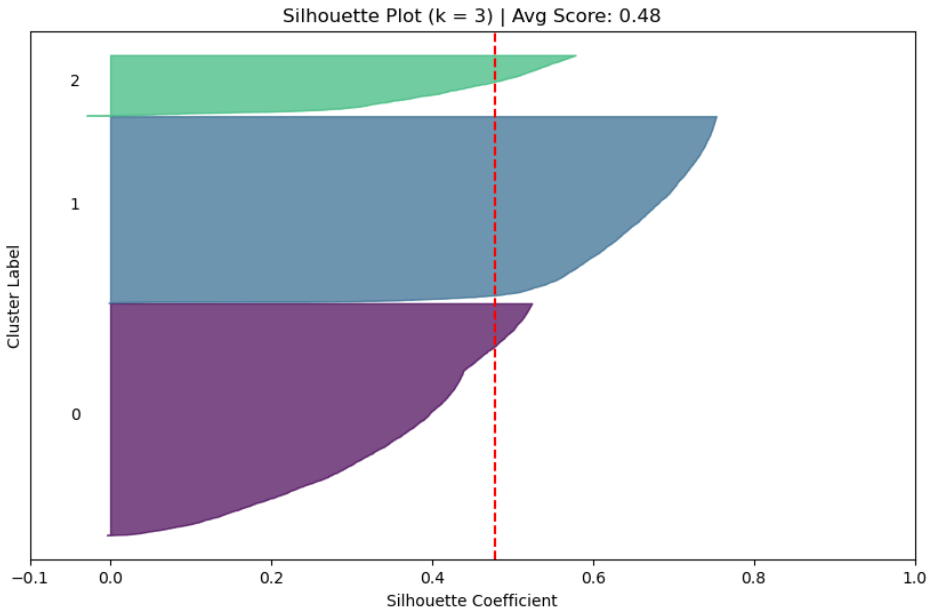

Figure: Silhouette Plot for KMeans Clustering (k = 3)

K=4:

The silhouette plot for k = 4 indicates moderate clustering performance, with most data points showing positive silhouette values. This suggests that the clusters are reasonably well-formed and separated.

However, some clusters display greater variability in silhouette scores, indicating slight overlap or less cohesive grouping compared to other clusters. Overall, while k = 4 provides acceptable clustering, the separation is not as strong across all clusters.

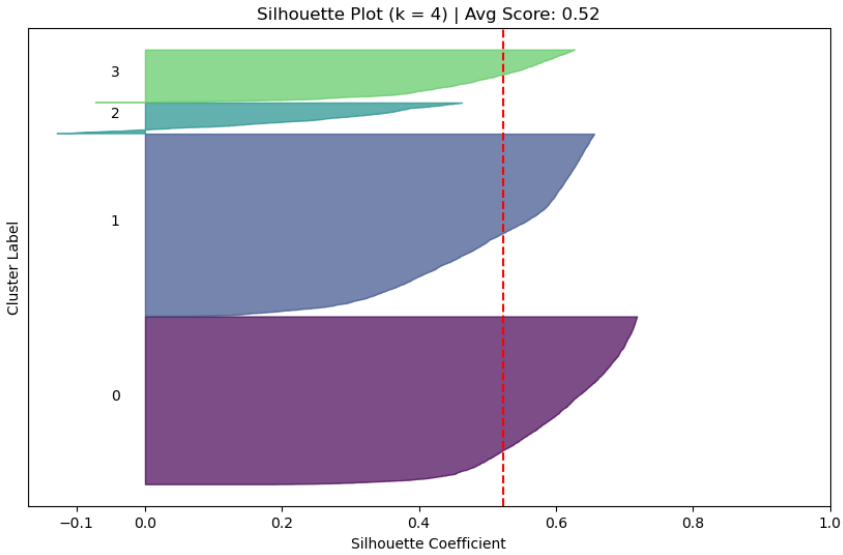

Figure: Silhouette Plot for KMeans Clustering (k = 4)

K=5:

The silhouette plot for k = 5 shows a moderate clustering structure, with most data points having positive silhouette values. However, compared to lower values of *k*, there is increased variability within clusters and some points with lower silhouette scores.

This suggests that the clusters are becoming less cohesive and more overlapping. Overall, while k = 5 provides reasonable segmentation, the cluster quality is slightly reduced compared to simpler models.

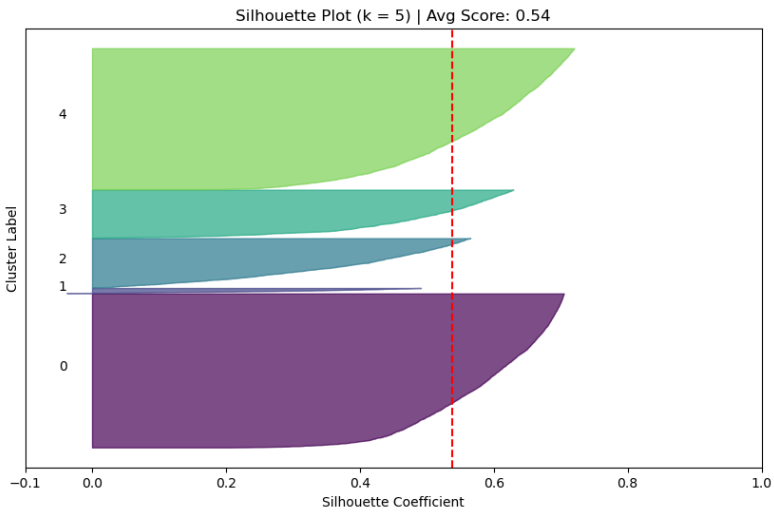

Figure: Silhouette Plot for KMeans Clustering (k = 5)

K=6:

The silhouette plot for k = 6 shows a moderate clustering structure, with most data points having positive silhouette values. However, compared to lower values of *k*, there is increased variability within clusters and some points with lower silhouette scores.

This suggests that the clusters are becoming less cohesive and more overlapping. Overall, while k = 6 provides reasonable segmentation, the cluster quality is slightly reduced compared to simpler models.

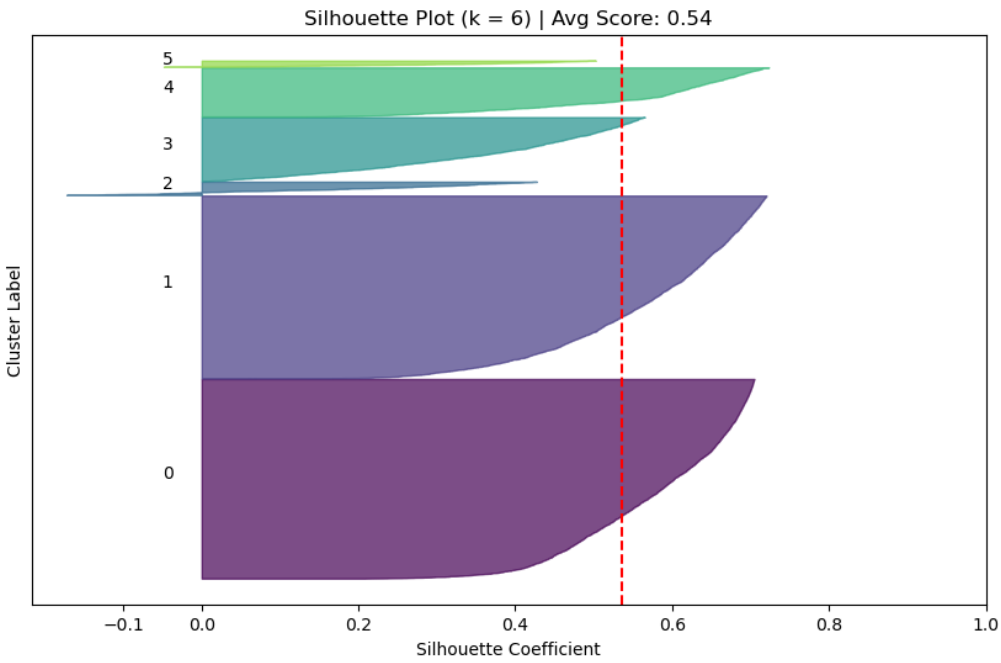

Figure: Silhouette Plot for KMeans Clustering (k = 6)

K=7:

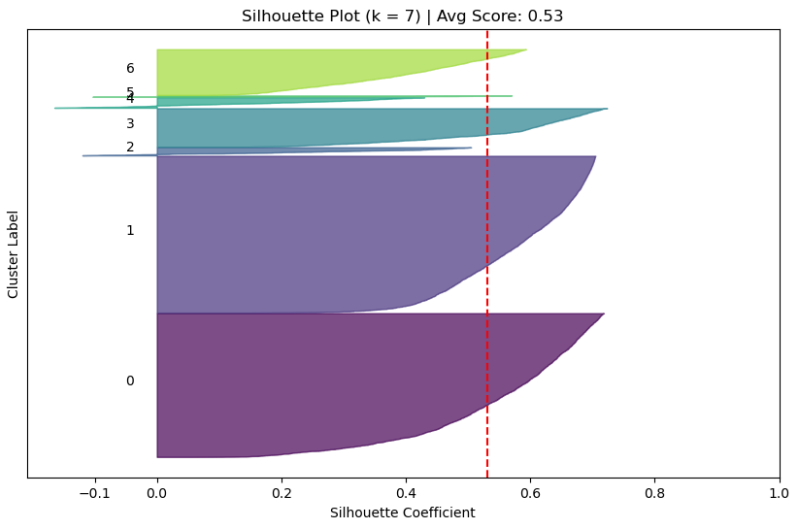

Figure: Silhouette Plot for KMeans Clustering (k = 7)

The silhouette plot for k = 7 indicates weaker clustering performance, with increased variability in silhouette values across clusters. Several data points show low or near-zero scores, suggesting overlapping clusters and less distinct group boundaries.

This implies that the data may be over-segmented at this level. Overall, k = 7 results in reduced cluster cohesion and separation, making it less suitable compared to lower values of *k*.

13.0 GMM Model Selection using BIC

The selection of an appropriate Gaussian Mixture Model (GMM) is a critical step in achieving effective clustering. In this study, the optimal GMM configuration is determined using the Bayesian Information Criterion (BIC), which evaluates model performance by balancing goodness of fit with model complexity.

	eps	min_samples	silhouette_score
0	0.1	2	-1.0
1	0.1	6	-1.0
2	0.1	10	-1.0
3	0.1	14	-1.0
4	0.1	18	-1.0

Table: DBSCAN Hyperparameter Tuning Results

Gaussian Mixture Model (GMM)

The Gaussian Mixture Model (GMM) is a probabilistic clustering technique used to model data as a combination of multiple Gaussian distributions. Unlike traditional clustering methods such as KMeans, which assign each data point to a single cluster, GMM estimates the probability of a data point belonging to each cluster. This allows for a more flexible and realistic representation of complex data structures.

Working Principle

GMM assumes that the dataset is generated from a mixture of several Gaussian distributions, each representing a cluster. The algorithm uses the Expectation-Maximization (EM) method to iteratively estimate the parameters of these distributions:

- **Expectation Step (E-step):**
  Calculates the probability that each data point belongs to each cluster
- **Maximization Step (M-step):**
  Updates the parameters (mean, covariance, and weights) of the Gaussian distributions

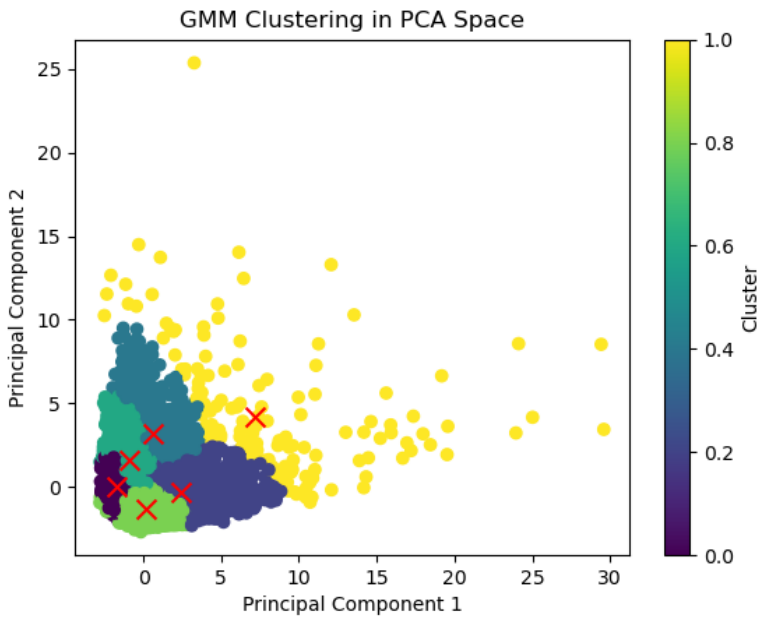

Figure: GMM Clustering Results in PCA Space

The GMM clustering results show overlapping clusters in the PCA space, indicating that customer groups are not clearly separated. This reflects the probabilistic nature of GMM, where data points can belong to multiple clusters with varying likelihoods.

Overall, the model captures more flexible and realistic patterns in customer behaviour, especially where cluster boundaries are not distinct.

DBSCAN Clustering

DBSCAN identifies clusters based on data density and detects noise points representing outliers. The results show less distinct cluster separation compared to KMeans and GMM, indicating limited effectiveness for this dataset.

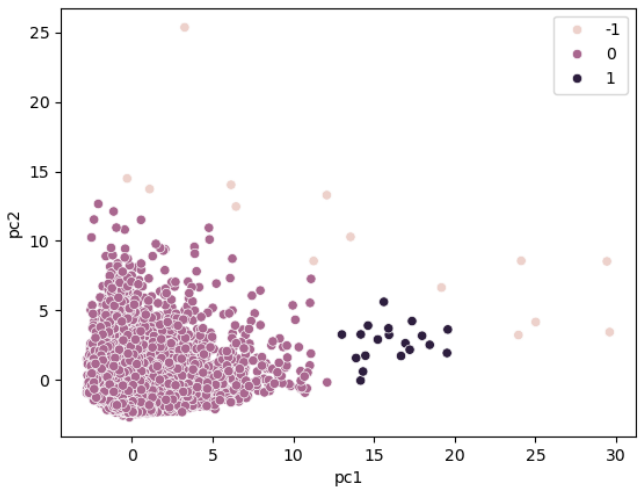

Figure: DBSCAN Clustering Results in PCA Space

KMeans Cluster Profiling

KMeans cluster profiling shows clear differences in feature averages across clusters, indicating distinct customer segments. These variations reflect differences in spending, credit usage, and transaction behaviour.

KMeans_pred_label	BALANCE	BALANCE_FREQUENCY	PURCHASES	ONEOFF_PURCHASES	INSTALLMENTS_PURCHASES	CASH_ADVANCE	PURCHASES_FREQUENCY	ON
0	1893.400240	0.979339	2916.566171	1986.694776	930.304921	353.408743	0.923747	
1	4541.393882	0.976638	15777.311395	10689.027791	5088.283605	1060.190695	0.928101	
2	4616.977320	0.967882	493.136111	302.052352	191.172014	4623.363303	0.280563	
3	645.955449	0.902331	797.744495	166.429623	631.809188	178.096535	0.856537	
4	1051.088979	0.797032	278.931482	225.848988	53.284815	619.569625	0.163421	

Table: KMeans Cluster Profiling Summary

Feature Distribution Across Clusters

The distribution of features across clusters highlights clear differences in customer behavior. Variations in spending, credit usage, and transaction patterns indicate that each cluster represents a distinct customer segment.

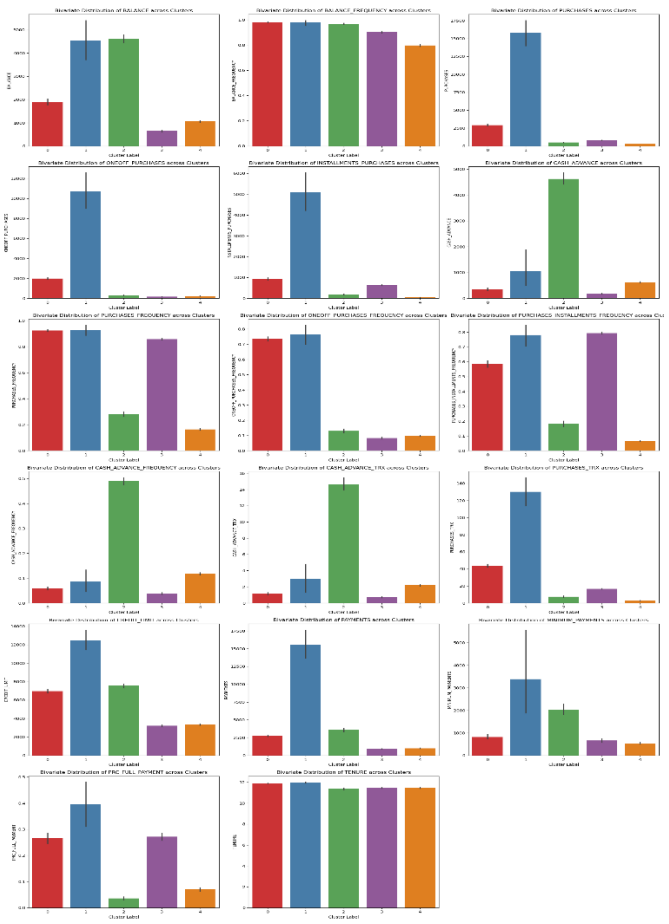

Fig: Feature Distribution Across KMeans Clusters

```
KMeans_stats = KMeans_stats.transpose()
KMeans_stats
```

KMeans_pred_label	0	1	2	3	4
BALANCE	1893.400240	4541.393882	4616.977320	645.955449	1051.088979
BALANCE_FREQUENCY	0.979339	0.976638	0.967882	0.902331	0.797032
PURCHASES	2916.566171	15777.311395	493.136111	797.744495	278.931482
ONEOFF_PURCHASES	1986.694776	10689.027791	302.052352	166.429623	225.848988
INSTALLMENTS_PURCHASES	930.304921	5088.283605	191.172014	631.809188	53.284815
CASH_ADVANCE	353.408743	1060.190695	4623.363303	178.096535	619.569625
PURCHASES_FREQUENCY	0.923747	0.928101	0.280563	0.856537	0.163421
ONEOFF_PURCHASES_FREQUENCY	0.734122	0.763090	0.130454	0.083195	0.096915
PURCHASES_INSTALLMENTS_FREQUENCY	0.585110	0.781501	0.182218	0.794342	0.065744
CASH_ADVANCE_FREQUENCY	0.058192	0.085271	0.490732	0.038047	0.118664
CASH_ADVANCE_TRX	1.154624	2.988372	14.649306	0.709611	2.205980
PURCHASES_TRX	43.521676	130.197674	7.368056	16.647059	2.951188
CREDIT_LIMIT	6972.541382	12493.023256	7547.920612	3199.132690	3342.325581
PAYMENTS	2776.468069	15581.496801	3589.972494	935.045366	1005.913641
MINIMUM_PAYMENTS	819.342437	3383.304083	2019.417524	662.860663	539.915352
PRC_FULL_PAYMENT	0.266129	0.394721	0.035889	0.271560	0.070684
TENURE	11.888006	11.965116	11.380208	11.454847	11.456683

Figure: KMeans Cluster-wise Feature Averages

13.0 Results and Conclusion:

The application of clustering techniques to the credit card dataset resulted in the identification of distinct customer segments based on financial behaviour. Among the models used, KMeans produced the most well-defined clusters, with clear separation and interpretability.

The analysis revealed multiple customer groups characterized by differences in spending patterns, transaction frequency, credit utilization, and cash advance behaviour. GMM provided additional insights by capturing overlapping patterns between clusters, while DBSCAN was effective in identifying outliers but did not form well-separated clusters.

Overall, the results demonstrate that the dataset contains meaningful structure, and clustering techniques can successfully uncover distinct customer segments for further analysis and decision-making.

Cluster 0 – High Spenders: Customers in this cluster exhibit the highest levels of purchases, transaction frequency, and credit limits. Their high repayment behaviour suggests strong financial capacity. These individuals are likely financially stable users who actively utilize their credit cards for large and frequent transactions.

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